POWER CABLE V2: Adaptive Resistance for a Compact Home Gym

Santiago Remis, Eusebio Rodriguez, Yunus Emre Sahin, Shaun Varghese

Abstract:

Home resistance training is limited by equipment that is often expensive, bulky, noisy, while being unable to adapt to the user's changing physical capacity. Fixed resistance requires users to have multiple weights and may not provide the appropriate training load as their strength and fatigue vary during exercise. This project develops Power Cable V2, a compact motorized cable system with an interface that measures user-applied force and enables automatic resistance selection based on the user's measured speed. The prototype achieved an average speed-measurement error of 8%, demonstrating accurate and consistent measurement for determining an appropriate training load, as well as a safer and more robust casing for the product.

1. INTRODUCTION

Home resistance training is attractive but restricted by space and cost: multi-gym stations need a large dedicated area, and equipment cost and limited home space are common participation barriers [1,2]. Fixed weights further constrain weightlifting since effective training relies on progressive overload, necessary for hypertrophy [3,4], often to momentary muscular failure [5]. The previous Power Cable project solved space and cost with a motorized cable system supplying calibrated electronic resistance, but its resistance was fixed at the start of each set, ruling out progressive overload and failure training. A user's force capacity, however, fluctuates daily with fitness, fatigue, and readiness [6]. Connected adaptive resistance devices that adjust resistance in real time have emerged as a solution [7], but commercial smart gyms remain expensive; the Power Cable prototype offered compact training at 4,000 SEK versus 28,300 SEK[5]. Our project develops an intelligent system that adapts resistance to the user's capacity based on their desired training speed.

2. DESIGN

The product was initially built by Linus Remahl [4] and later worked on by a group of previous students [1].

2.1. Hardware

The hardware improvements made to the Power Cable previous prototype involved adding a cable box to enclose the power supply, Odrive controller, and Bluetooth antenna with an MDF box to avoid safety hazards related to cables coming from the power supply to the motor being exposed to the environment.

The box was designed using the open-source *boxes.py* tool and Inkscape to draw the cable holes. The dimensions were: 240x130x110 millimeters, and the cable holes in the front of the box are 40mm in diameter, and the back hole is 70 millimeters in diameter. When the design was finished, the shape was exported as an SVG file. Then we used a CNC machine to cut the enclosure out of 6-millimetre MDF.

Figure 1 shows the prototype together with the mechanical CAD design by the previous group. This figure emphasizes the accessibilities of the components such as the ODrive controller and wiring.

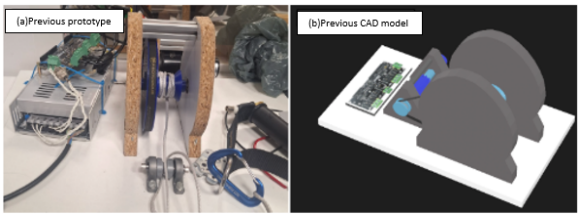


Figure 1) Previous prototype: (a) photograph taken by group member and (b) CAD-model developed by the previous project group [5]

Figure 2 shows how the motor is operated, the software interfaces send the commands to the Odrive board, which tells the motor encoder what is the current limit it can use to try to get back in place and give resistance to the user.

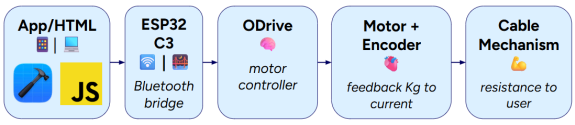


Figure 2) Motor control workflow

2.2. Software

The Power Cable V2 project provides several software methods of controlling the device. These include a browser-based serial interface, an iOS application using Bluetooth, and a wireless web interface hosted directly by the ESP32-C3.

2.2.1 Browser-Based Control Interface

The existing Python functions were transferred into an HTML interface that has been developed to operate the device directly from the GitHub Repository "README" file. From this markdown file the user is able to operate the device with any computer regardless of the operating system and get information about the project. The page communicates with the ODrive through a USB and runs without Python packages, but it still requires a compatible browser and the serial adapter [4].

Github Page:

amsimer.github.io/Weightlifting-project/

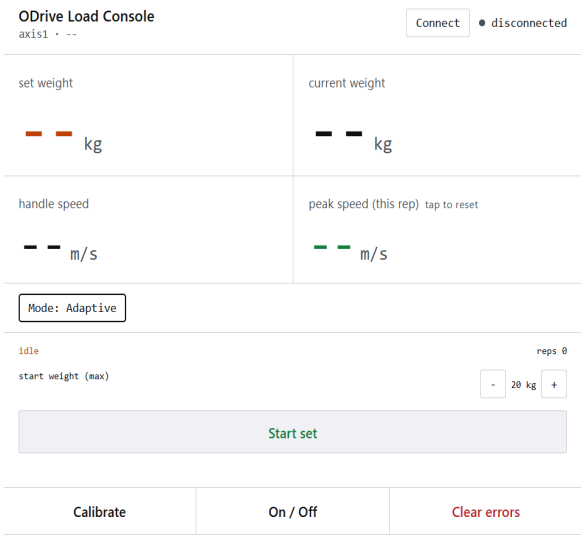


Figure 3) HTML software control interface

The interface allows the user to calibrate the device, set a starting position, switch between fixed weight and adaptive weight modes, and turn the device on and off. In the fixed mode, the user can select a fixed training weight with and look at his current pulling force (in kg) and his current speed and previous rep peak speed. In the adaptive mode, the user does a calibration rep at the speed they desire to train, and then each rep starts off heavy and goes down until the user is able to do the repetition at their desired speed.

2.2.2 Mobile Application

An iOS application was developed to provide portable control of the Power Cable V2. The application communicates with the ESP32-C3 through Bluetooth. The ESP32-C3 receives commands from the application and forwards the required instructions to the ODrive motor controller through UART.

The application allows the user to calibrate the motor, select the training resistance, configure the target number of repetitions and sets, and start or stop a training session. During exercise, it displays the current repetition count, set number, cable speed, set duration, and feedback message received from the controller. Audible feedback informs the user about repetition count and the resistance increases or decreases in the next set depending on the user's speeds.

2.2.3 Wireless Web Interface

A second interface was developed to make the system accessible without installing the iOS application or connecting the device through USB. The ESP32-C3 creates a local Wi-Fi access point named POWER-CABLE and hosts the website directly from its LittleFS storage. After connecting to this network, the user can open the interface at the local address 192.168.4.1, shared on GitHub. An internet connection is not required.

The training dashboard displays live cable speed, repetition progress, set duration, completed repetitions, and measurements from the previous repetition. Cable speed is also presented in a live graph. After each set, the interface calculates the velocity loss and displays a recommended resistance for the next set.

Figure 4 displays the wireless user interfaces. The iOS application and wireless website provide two independent interfaces for the same training system. Both communicate through the ESP32-C3 firmware and use the same motor-control and safety functions.

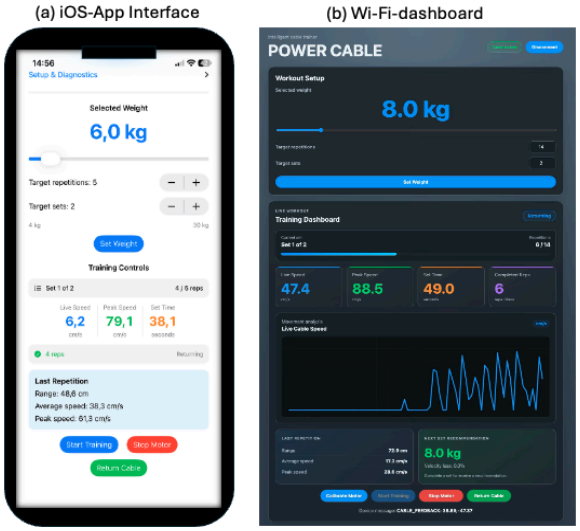


Figure 4) Wireless User Interfaces: (a) iOS-App Interface and WiFi-dashboard

3. RESULTS

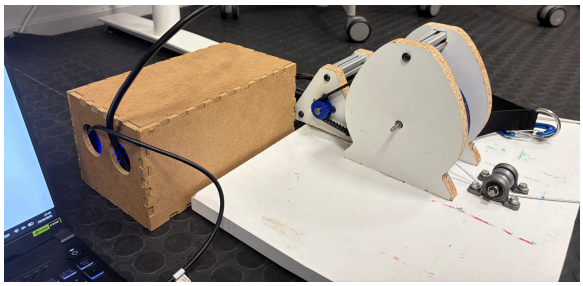


Figure 5) Final prototype of Power Cable V2.

The power supply unit and the ODrive are enclosed and have an opening cover for easy use of tools. The improved design includes the addition of a cable box to cover the power supply, ODrive, wires, bluetooth antenna and circuit boards. The device uses current to try to get into position after the user pulls the cable, the max current determines the force it uses to get back. This current is calculated from the desired weight in Kg with this formula: I = Kg/1.8.

To validate the speed measuring, which is used in the adaptive resistance mode, the speed was measured with the video tracking software: Tracker. The measured speed by the interface and Tracker were compared for 5 different repetitions.

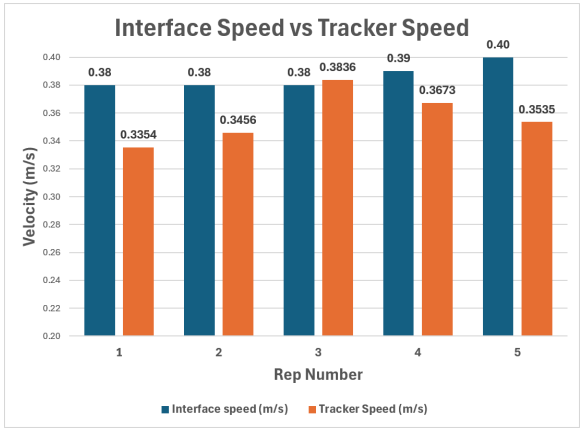


Figure 6) Real vs measured velocity comparison

The speed of both methods were close on each repetition with an average error of 8%. This is deemed to be acceptable for measuring the user's rep speed and determining the optimal resistance value. To further increase the accuracy, the use of external speed sensors could be implemented on the design.

4. CONCLUSION

The project demonstrated that the previous Power Cable prototype could be developed into a more accessible and robust adaptive resistance-training system. The new enclosure protects the power supply, ODrive controller, wiring, and the circuit boards while still allowing the training cable to operate normally. The control system was extended with three user interfaces: a USB-based browser interface, an iOS application, and a wireless dashboard. The app and wireless dashboard allow the user to operate the device in the most portable way, while tracking various training data, providing audible feedback and recommending resistance adjustments based on the measured velocity loss. The HTML interface removed the need of installing python, being accessible directly from the github. It also incorporates the adaptive resistance mode based on the user's training speed, which was validated through comparisons with video tracking and gave an average error of 8%. Further work should evaluate the system with more users and exercises, improve wireless reliability, and validate resistance accuracy and safety under standardised test conditions.

REFERENCES

[1] Persson, H.; Edwall, L.; Alhadeff, C.; Fabricius-Hansen, A.; Yasar, M. Power Cable: Effortless Weight Selection for Compact, Economical Home Gyms. (Built on Remahl's original machine, https://github.com/linusreM)

[2] Kassiano, W. et al. Progressive Overload Affects the Magnitude of Muscle Hypertrophy. Medicine & Science in Sports & Exercise, 2026.

[3] Plotkin, D. et al. Progressive overload without progressing load? The effects of load or repetition progression on muscular adaptations. PeerJ, 2022.

[4] Linus Remahl: https://github.com/linusreM

[5] hugo3395: https://discord.com/channels/1541729194508288030/1541873862701224077/1550827046392045619

[6] Gaspar, M.; Ventura, F.; Pereira, C.; Santos, C. M. Continuous Fitness at Home: Designing Exercise Equipment for the Daily Routine. 2006.

[7] Rosaini, N. D.; Gray, S. R.; Oyebola, E.; Ryde, G. C. How women engage in muscle-strengthening exercises: a qualitative study from the WISH project. BMC Public Health, 2026.